OBJCTIVES OF CURRENT PAPER

ZOO501- DEVELOPMENT BIOLOGY

Final-Term File 2023 By

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1. A condition in which cell lacks instructions for development is **Naïve.**
2. The size of Dna in nucleosome is almost? **140 bp**
3. How many responses and egg delivered to sperm during fusion process? **2**
4. which of the following progress through a cell cycle consisting of just two steps? **Blastomeres**
5. The ability of the embryonic cells to change their fates to compensate for the missing is called **regulation**
6. when potassium ions leave the sperm then it's membrane potential become? **Negative**
7. A membrane that surrounds bones is called **periosteum.**
8. The major cells adhesion molecules appear to the **cadherians.**
9. which type of neurulation is common in fishes ? **Secondary neurulation**
10. In rana pipiens egg development last **3 years**
11. major classes of cadhrin includes **N cadherians**.
12. florescent dyes such as fura 2 is used to visualized which of the following? **Calcium**
13. axoneme is formed by **microtubules..**
14. Most Transcription factors can bind to specific **DNA**
15. fusion of **egg** And sperm cell plasma membrane.
16. Egg plasma membrane changed the potential by changing the concentration **sodium.**
17. calcium and bicarbonate ions may be critical in activating **cAMP?**
18. This resting membrane potential is generally about **70mV**
19. Zp3 is the specific glycoprotein in the **mouth.**
20. The acrosomal reaction in sea urchins Is initiated by contact of the sperm and with the **jelly.**

21. Finally ,a fourth cortical granule protein hyaline forms a coating around the **egg.**
22. strateging for establishment commitment are **2**
23. the mature sea urchin egg is a metabolically sugglish cell that is activated by the **sperm.**
24. during the process of development division of embryo leads to how many layers? **Three**
25. the opening of archerteron at the vegetal region is called **blastopore.**
26. fusion of the genetic material of sperm and egg is called **fertilization.**
27. the orientation of cleavage plane is snail eggs controlled by **cytoplasmic factors.**
28. the blastomere cell cycle is governed by the synthesis degration of **cycle**
29. in tunicate cleavage at the **32** stage cell a small blastocoel.
30. The fetal portion of placenta formed by _____ **Chorion**
31. The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the area **Opaca.**
32. the rudiment of the central nervous system, is called **Neurulation.**
33. Archenteron is used for **Primitive Gut**
34. Spiral holoblastic cleavage is characteristic of several animal groups including **annelid,**worms, some flatworm.
35. In sea urchins, cell fates are determined by **signaling.**
36. The ype of cell forms the **syncytiotrophoblas**
37. The cells at the dorsalmost portion of the neural Tube become the **neural crest cells.**
38. The Myelencephalon eventually becomes the **Medulla oblongata**
39. One agent .the human posterior neural tube regions **at day 27**. **(close)**
40. Failure to close the anterior neural tube regions results in a lethal condition **anencephaly**
41. original ependymal layer of the cerebellum generates a wide variety of neurons and glial cells, including the distinctive and large **Purkinje neurons.**
42. TGF-β Family secreted from the surface ectoderm and roof of the **neural tube.**
43. Paracrine factors instruct the ……cells to become muscles. **(myotome**
44. The osteoblasts secrete a colondral ossification involves the formation of **cartilage tissue**
45. Neuive groove **depression**
46. Vegetal pole :**Dalton protein , 30500 Da**
47. **Primary neurolation**
48. Identical twins….**0.25%**
49. Acadians have notochord but lack **vertebrae**
50. Neural crest formee from.. portion most dorsal **Area opeca**
51. Sub germinal cavity Mantle zone have neurons ..**gilia**
52. Cluster of neurons….**nuclei**
53. **poblast**……cell determine inner **cell Mass**
54. Contracting long thin (30×5 um) …..**filopodia**
55. Perechordal **plate** induc…**forebrain**
56. Fertile portion is **chorion**
57. **Tgf** and fgf7 …in skin development
58. Teleo lecithal egg include **birds fish**
59. Sox9 gene….**encode DNA**
60. Direct conversion of mesenchymal tussu into bones ……. **Intramembrane ossification**
61. In see urchins fate determine by …..**signaling**
62. To …… anterior neural tube region anencephaly .**close**
63. Mesencephalon become. **Medulla oblongata**
64. Spiral holoblastic cleveage in…….. worms flatworms molluscs **annelids**
65. Which of the following insect follow all three modes of specification--- **Drosophila**

66. Silencers are molecules which are basically ---- **negative enhancers**
67. The sperm travel by the amoeboid motion of ------- of the cell membrane. _____**Lamellipodial Extensions**
68. The mechanical agent of cytokinesis is a contractile ring of microfilaments made of **actin**
69. These cells begin extending and contracting long, thin (30 × 5 μm) processes called **filopodia** from their inner surfaces.
70. In the typical cleavage of **molluscs**, cells divides to produce most of the larval structures
71. They have a notochord as larvae (and therefore are chordates), but they lack **vertebrae**.
72. The presumptive notochord cells on the right and left sides of the **embryo migrate to the midline and interdigitate to form the notochord**, a single row of **40** cells
73. Cyclin synthesis promotes the formation of **MPF**, and MPF promotes mitosis.
74. The first **12** divisions occur synchronously, forming a mound of cells that sits at the animal pole of a large yolk cell.
75. Fertilization of the chick egg occurs in the **oviduct**, before the albumen and the shell are secreted upon it.
76. The major structural characteristic of avian, reptilian, and mammalian gastrulation is the **primitive streak.**
77. This fetal organ the **Chorion** is derived primarily from embryonic trophoblast cells.
78. The original type of **trophoblast** cells constitute a layer called the cytotrophoblast.
79. The prechordal plate induces the formation of the **forebrain.**
80. In birds **pH** differences appear crucial for distinguishing dorsal from ventral.
81. The rudiment of the central nervous system, is called **neurulation**
82. The cells at the dorsalmost portion of the neural tube become the **neural crest cells.**
83. The most crucial, difference between mammalian cleavage and all other types involves the phenomenon of_____ **Compaction**
84. Between the area pellucida and the area opaca is a thin layer of cells called the _____**Marginal Zone (Or Marginal Belt).**
85. The telencephalon will eventually form the cerebral hemispheres, and the diencephalon will form the thalamic and -------brain regions_____ **Hypothalamic**
86. TGF-β family secreted from the surface ----- and roof of the neural tube. ____ **Ectoderm**
87. -----is characteristic of several animal groups including annelid, worms, some flatworms, and most mollusks_____ **Spiral holoblastic cleavage**
88. In sea urchins, cell fates are determined by_____ **Signaling**
89. The ..... constitute a major signaling center_____ **micromeres**
90. Shell formation___ **snails**
91. Multinucleated type of cell forms the_____ **syncytiotrophoblast**
92. The cells at the dorsal most portion of the neural tube become____ **neural crest cells.**
93. The Myelencephalon eventually becomes the____ **Medulla oblongata**
94. One agent of ventral specification is the____ **sonic hedgehog.**
95. Chlamydomona usually haploid having---- Copies of chromosomes___ **Only one**
96. Pdx1 is specific for------ region of endoderm____ **pancreatic**
97. Protein synthesis is accomplished by ribosomes and_____ **tRNA**
98. Sea urchins exhibit which cleavage____ **radial holoblastic**
99. Mouse fertilin is localized to the posterior plasma membrane of the sperm____ **head**
100. Mesoderm forms____ **Kidneys**
101. MPF have how many units_____ 2
102. The major cell adhesion------ appear to be the Cadherins____ **Molecules**

103. The human β-globin gene has an enhancer in its 3′ UTR, roughly .....base pairs___ **700**

104. The mammalian egg is also surrounded by a layer of cells called____ **cumulus**

105. The fetal portion of placenta formed by ___ **Chorion**

106. The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the area ___ **Opaca.**

107. The rudiment of the central nervous system is called____ **Neurulation.**

108. Archenteron is used for _____**Primitive Gut**

109. Failure to------ the human posterior neural tube regions at day 27___ **close**

110. Failure to close the anterior neural tube regions results in a lethal condition____**anencephaly**

111. Originnal ependymal layer of the cerebellum generates a wide variety of neurons and glial cells, including the distinctive and large___ **Purkinje neurons.**

112. TGF-β family secreted from the surface--- and roof of the neural tube____ **ectoderm**

113. Paracrine factors instruct the .....cells to become muscles____ myotome

114. In MPF Larger subunit is called____ **B cyclin**

115. Cytokinesis is the division of____ **Cytoplasm**

116. These cells begin extending and contracting long, thin (30 × 5 μm) processes called____ **filopodia.**

117. Ascidians have_____ **Notochord**

118. β-catenin is important for the inducing capacity of the._____ **micromeres**

119. Since only the cytoplasm of the blastodisc becomes the embryo, this type of meroblastic cleavage is called____ **discoidal**

120. The center of Henson's node contains a funnel-shaped depression called____ **primitive pit.**

121. Endochondral ossification involves the formation of_____ **cartilage tissue**

122. Neurulation in----- is exclusively secondary_____ **Fishes**

123. The remaining identical twins share a common_____ **chorion**

124. The vegetal (neck) neural crest lies opposite chick somits_____ **1-7**

125. The direct conversion of mesenchymal tissues into bone is called_____ **intramembranous ossification.**

126. Gastrulation in frog begins at a point on the embryo surface roughly____ **180 degree**

127. In humans the SOX9 gene which----- a DNA binding protein_____ **encodes**

128. Originnal ependymal layer of the cerebellum generates a wide variety of----and glial cells____ **neurons**

129. Congenital anomalies can be caused by genetic factors and__ **environmental agents**

130. The most crucial, difference between mammalian cleavage and all other types involves____ **Compaction**

131. In mammals after ovulation from ovary ovum is captured by____ **Fimbriae**

132. Identical twins occur in roughly------of human births_____ **0.25%**

133. About ------- of identical twins have two complete and separate chorions_____**33%**

134. The first cell movement of fish gastrulation is the --------of the blastoderm cells over the yolk_____ **Epiboly**

135. During a process called------the trophoblast cells secrete fluid into the morula to create a blastocoel_____ **cavitation,**

136. The cell divisions do not completely divide the egg, so this type of cleavage is called_____ **meroblastic.**

137. Protocadherins have been found to be very important in separating the notochord from the other mesodermal tissues during--gastrulation_____ **Xenopus**

138. The original type of trophoblast cells constitute a layer called the____ **cytotrophoblast**

139. Slow block, initiated by the intracellular release of_____ **Calcium Ions**

140. A condition in which cell lack instructions for development is called _____**Naive**

141. During fertilization -------- molecules fuse to form long microfilaments. **Actin**

142. Major class of cadherin includes _____**N- Cadherin**

143. The osteoblasts secrete a collagen-proteoglycan -- that is able to bind calcium salts. __**Matrix**

144. The second step is the ------of the myoblasts together into chains_____ **Alignment**

145. In frogs and chicks, ------- neurulation is usually seen in the neural tube ____**Secondary**

146. . The mantle zone cells differentiate into both _____**Neurons and Glia**

147. Mother provides the fetus with nutrients and oxygen, and the fetus sends its waste products---- into the maternal circulation_____ **mainly carbon dioxide and urea**

148. Paracrine factors such as TGF- and FGF7 are important in normal ___**Skin Development.**

149. Dendrites are often numerous and do not extend far from the neuronal cell body, or------ axons may extend for several feet. _____ **Soma**

150. the space between two neurons which transmit are released_____ **Synapse**

151. Paracrine factors such as --- and FGF7 are important in normal skin development. ____ **TGF**

152. The depression is called the _____**Primitive Groove**

153. Fishes, ------, and birds undergo discoidal meroblastic cleavage, wherein the early cell divisions do not cut through the yolk of the egg____ **Reptiles**

154. Gastrulation in Snails is accomplished primarily by_____ **Epiboly**

155. They have a notochord as larvae but they lack _____**Vertebrae**

156. Fertilization of the chick egg occurs in the _____**Oviduct,**

157. Transcription factors have---- major domains_____ **three**

158. Mutations of the -------- cause campomelic dysplasia, a rare disorder of skeletal

159. development that results in deformities of most of the bones of the body_____**SOX9 Gene**

160. Mutation of sox9 gene cause _____**Campomelic Dysplasia**

161. In birds -------differences appear crucial for distinguishing dorsal from ventral. _____**pH**

162. The second step of gastrulation involves the_____ **Involution of The Mesoderm.**

163. The mammalian oocyte is released from the ovary and swept by the fimbriae into_____**the Oviduct.**

164. It secretes a ----- tunic (which gave the name "tunicates" to these creatures) ____**Cellulose**

165. The second step of gastrulation involves the…….. **Involution of The Mesoderm**

166. constitute a major signaling center……**Micromeres**
167. The cytoplasm of the blastodisc becomes the embryo, this type of meroblastic cleavage is called…….**Discoidal**
168. The yolk syncytial layer is formed at the____ when the cells at the vegetal edge of the blastoderm fuse with the underlying yolk cell…….**9th cell cycle**
169. Fertilization of the chick egg occurs in the……**Oviduct**
170. The center of this node contains a funnel-shaped depression through which cells can pass into the blastocoels called as…… **Primitive pit**
171. During mammalian cleavage the trophoblast cells secrete fluid into the morula to create a blastocoels this process called as……..**Cavitation**
172. Neurulation in _____ is exclusively secondary ……**Fishes**
173. Failure to __the human posterior neural tube regions at day 27…. **Close**
174. Fertilization occurs in the -------- of the oviduct, a region close to the ovary___ **Ampulla**
175. The major structural characteristic of avian, reptilian, and mammalian gastrulation is the_____ **Primitive Streak.**
176. A condition in which cell lack the instructions for development **naïve**
177. Major classes of cadherin include……., **N-cadherin**
178. Spiral cleavage found in ……**Snails**
179. Spiral holoblastic is characteristic of several animals 'groups, including_____, some flatworms, and most mollusks (snails)……. **Annelid worms**
180. They have a notochord as larvae but they lack …….**Vertebrae**
181. The Myelencephalon eventually becomes the……. **Medulla oblongata**
182. One agent of ventral specification is the ___hedgehog protein… **Sonic**
183. The mantle zone cells differentiate into both …….**Neurons and Glia**
184. Meanwhile, the original ependymal layer of the cerebellum generates a wide variety of neurons and glial cells, including the distinctive and large……. **Purkinje neurons.**
185. Chlamydomonas Which is usually haploid, having just ……**one copy of each chromosome.**
186. The mammalian egg is also surrounded by a layer of cells called the
187. ……..**cumulus**
188. The major structural characteristic of avian, reptilian, and mammalian
189. gastrulation is the……..**Primitive Streak.**
190. The vagal (neck) neural crest lies opposite chick somites……. **1-7**
191. Paracrine factors instruct the_____to become muscles by inducing them to synthesize the Myoid protein……..**Myotome Cells**
192. Since only the cytoplasm of the blastodisc becomes the embryo, this type of meroblastic Cleavage is called……. **Discoidal**
193. Controlling the Efficiency and rate of transcription from……. **Enhancer**
194. The growing region at the ____of the lip is called the chordoneural
195. hinge……**tip**
196. The mammalian oocyte is released from the ovary and swept by the fimbriae into……..**The Oviduct.**
197. Which portion form in small yolk…….. **animal pole**
198. The cells at dorsal most portion of neural tube become ….**Neural Crest Cells.**
199. The cerebral cortex in humans has ____ layers…… **6**

200. The second step is the ____of the myoblasts together into chains…….
Alignment

201. J.Hammerling exchanged nuclei b/w two _____distinct species……**morphologically**